

Experiment 12: Motion Segmentation in Video

Theory:

Motion segmentation is the process of separating moving objects from the background in video sequences. It plays a crucial role in applications such as traffic surveillance, human activity recognition, and video compression. A common approach involves calculating the difference between consecutive frames to identify regions where motion has occurred, followed by noise reduction and contour detection to localize moving objects.

Algorithm:

1. Load the input video and read two initial frames.
2. Compute the absolute difference between the frames to detect changes.
3. Convert the difference to grayscale for simplification.
4. Apply Gaussian blur to smooth out noise.
5. Apply thresholding to extract prominent motion regions.
6. Detect contours in the thresholded image.
7. Filter out small contours and draw bounding boxes for larger moving regions.
8. Write the processed frames to an output video.
9. Repeat the process until the end of the video.
10. Release video capture and writer resources.

Flowchart (written):

Start → Load video → Read two frames → Compute absolute difference → Convert to grayscale → Apply Gaussian blur → Apply threshold → Find contours → Filter small contours → Draw bounding boxes → Write frame to output → Read next frame → Repeat until done → Release resources → End

```
import cv2

# Open the video
cap = cv2.VideoCapture('traffic.mp4')

# Get video properties
frame_width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
frame_height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
fps = cap.get(cv2.CAP_PROP_FPS)

# Define the codec and create VideoWriter object
out = cv2.VideoWriter('output_motion.avi', cv2.VideoWriter_fourcc(*'XVID'), fps, (frame_width, frame_height))

# Read two initial frames
ret, frame1 = cap.read()
ret, frame2 = cap.read()

while cap.isOpened():
    diff = cv2.absdiff(frame1, frame2)
    gray = cv2.cvtColor(diff, cv2.COLOR_BGR2GRAY)
    gray = cv2.GaussianBlur(gray, (5, 5), 0)
    _, thresh = cv2.threshold(gray, 25, 255, cv2.THRESH_BINARY)
    contours, _ = cv2.findContours(thresh, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)

    for contour in contours:
        if cv2.contourArea(contour) < 500:
            continue
        (x, y, w, h) = cv2.boundingRect(contour)
        cv2.rectangle(frame1, (x, y), (x+w, y+h), (0, 255, 0), 2)

    display = cv2.resize(frame1, (800, 600))
    cv2.imshow("Motion Segmentation", display)
    out.write(frame1)

    frame1 = frame2
    ret, frame2 = cap.read()
    if not ret or cv2.waitKey(30) & 0xFF == ord('q'):
```

```
break
```

```
cap.release()  
out.release()  
cv2.destroyAllWindows()
```